

SOM DEV BISHOYI

CONTACT INFORMATION

University of Massachusetts Dartmouth
Department of Mathematics
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EDUCATION

University of Massachusetts Dartmouth

Ph.D. in Computational Science and Engineering,
Specialization in Numerical PDEs, Scientific ML,
Gravitational waves, GPA : 4.00/4.00
Advisors : Dr. Scott Field, Prof. Gaurav Khanna

Sept 2022-present

IISER Kolkata

BS-MS in Physics, Minor in Mathematics,
GPA : 9.05/10

August 2017-July 2022

Thesis : [Studies on the spacetime of slowly rotating stars](#)

RESEARCH INTERESTS

I develop efficient, high-order PDE solving algorithms for problems in computational general relativity, black hole perturbation theory(BHPT) and gravitational waves. I work with discontinuous Galerkin/WENO methods to solve wave equations in BHPT. Also developing exact nonreflecting boundary conditions using Boundary Kernels and perfectly matched layers/hyperboloidal slicing.

PUBLICATIONS & PREPRINTS

1. Som Dev Bishoyi, Scott Field, Stephen Lau, *Radiation outer boundary conditions and near-to-far field signal transformations for the Bardeen-Press equation*, [arXiv:2604.22734](#).
2. Som Dev Bishoyi, Subir Sabharwal, Gaurav Khanna, *Source-Driven Tails in Kerr Spacetime : Nonlinear effects in Late-Time Behavior*, [arXiv:2601.11484v1](#)
3. Som Dev Bishoyi, Subir Sabharwal, Gaurav Khanna, *Numerical Evidence for Non-Axisymmetric Gravitational “Hair” for Extremal Kerr Black Hole Spacetimes with Hyperboloidal Foliations*, [10.1007/s10714-025-03378-1](#).

CURRENT PROJECTS

ROBCs and Near-to-Far field teleportation for the Teukolsky equation

- Development of boundary kernels to construct exact non-reflecting BCs to numerically solve the Teukolsky equation in Kerr spacetime.
- Development of teleportation kernels that perform near to far field teleportation of signals and extract them in the astrophysical wave zone.
- Above techniques enable long and stable simulations of Teukolsky equation in Boyer-Lindquist coordinates to Kerr spacetime in time domain.

General spin-weight time domain Teukolsky equation solver

- Discontinuous Galerkin(DG) solver for gravitational perturbations in the time domain having spectral convergence.
- Computing the Teukolsky source term for gravitational perturbations.
- Formulating a symmetric hyperbolic system of coupled PDEs using different choices of auxiliary variables and hyperboloidal coordinates.
- Implemented hyperboloidal slices (a variant of the perfectly matched layers technique) in the DG method to evolve wave equations (Bardeen-Press-Teukolsky equation).

AWARDS

- **UMass Dartmouth Doctoral Fellowship**, Research fellowship for a period of 1 year. *Sept 2022 to May 2023*
- **UMass Dartmouth Provost Travel Grant**, Travel grant of \$500 for presenting at conferences. *APS April Meetings 2024, 2025, 2026*
- **APS DGRAV Travel Grant**, Travel grant of \$300 for presenting at APS april meeting 2024. *April 3 2024 - April 6 2024*
- **IISER-K Summer Fellowship** Fellowship of 10,000 rupees for research project on Aharonov-Bohm effect and geometric phases. *May 2019 - July 2019*
- **IIT Indore Research Internship** Internship for two months on cosmological N Body Simulations. *May 2020 - July 2020*

INVITED TALKS

1. [Scalar and gravitational horizon hair as observable imprints of extremal black holes](#), *Extremal Black Holes and Black hole thermodynamics workshop, ICERM, Brown University, Jan-2026*

CONTRIBUTED TALKS AND POSTERS

1. Radiation Outer Boundary Conditions for the Teukolsky equation, *Eastern Gravity Meeting, Cornell University, May 2026*
2. Radiation Outer Boundary Conditions for the Teukolsky equation, *SXS NR Community Call, April 2026*
3. Radiation Outer Boundary Conditions for the Teukolsky equation, *APS Global Physics Summit, March 2026*
4. Exact boundary conditions for the Teukolsky equation, *Prospects in Theoretical Physics workshop, Institute of Advanced Study, Princeton, July 2025*
5. Exact boundary conditions for the Teukolsky equation, *SciML for gravitational wave astronomy workshop, ICERM, Brown University, June 2025*
6. Long simulations of extreme mass ratio inspirals by solving the Teukolsky equation with singular source terms, *APS Global Physics Summit 2025*
7. Determining extremality of Reissner-Nordstrom BHs using late time tails at $\mathcal{I}^+$, *APS April meeting 2024*

TEACHING EXPERIENCE

Department of Mathematics, UMass Dartmouth

- [TA for High Performance Scientific Computing](#) *Sept 2024 to Dec 2024*

Department of Physical Sciences, IISER Kolkata

- TA for Classical Mechanics II *August 2020 to Dec 2020*
- TA for Classical Mechanics II *August 2021 to Dec 2021*
- TA for Introductory Electromagnetism *January 2021 to May 2021*

Computation and Data Sciences, IISER Kolkata

- TA for Scientific Computing in Python *January 2022 to May 2022*

ADDITIONAL
EXPERTISE

Computing : C, Python(scipy, numpy, sympy, JAX), numerical PDE solvers, pseudo-spectral methods, finite difference methods, Mathematica xAct, git

Organizing : Biweekly talks at meetings of gravity research groups of UMass Dartmouth and University of Rhode Island(U2GRC).

Research Mentorship :

Project : **WENO solver for the massive Klein-Gordon equation**

- Varenja Upadhaya, 2nd year PhD student at UMass Dartmouth, Spring 2025

Project : **DG solver for wave equations in modified gravity spacetimes**

- Scott Shaw, MS-Physics student at UMass Dartmouth, Spring 2023
- Ansh Gupta, MS-Physics student at IISER Mohali, 2023

Referee for Physical Review D(2026)

PROFESSIONAL
MEMBERSHIPS

1. Simulating eXtreme Spacetimes (SXS)
2. LISA consortium
3. UMass-URI Gravity Research Consortium (U2GRC)
4. American Physical Society (APS)
5. Society for Industrial and Applied Mathematics (SIAM)